Orbital Periods for K2 Single-transit and Single-eclipse Events from K2 Revisits and TESS

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Abstract

LaCourse 2018 catalogued 164 K2 events seen only once. We searched all later K2 campaigns and all TESS data for repeats of the 144 events that TESS should detect. Every period we report comes from fitted eclipse times and was re-derived blind with independent code. We link K2 C5 and C16 transits to the TESS transits of TOI-5560.01, narrowing its period from ≤702 d to 140.4939 or 70.2470 d. The K2 planet candidate EPIC 211503363 has P=122.905 or 61.453 d, but its companion is ≈2.2 R_J, probably an M dwarf. We give new or refined periods or short alias lists for nine eclipsing binaries. These include HD 220967, a bright, eccentric (e≥0.58), 73.245-d system. In another, EPIC 202126877, the primary eclipse deepened from 6% in 2014 to 20% in 2021 and 32% in 2025.

Keywords: Exoplanet detection methods (489); Transit photometry (1709); Eclipsing binary stars (444)

Sample and Methods

The catalogue lists 164 events on 163 stars. We kept the 144 events that have TESS coverage, an expected TESS signal-to-noise ratio of at least 7 per event, and a physical depth. For each star we combined K2 PDCSAP light curves from every campaign (Howell 2014) with TESS SPOC, TESS-SPOC (Caldwell 2020) and QLP (Huang 2020) light curves. For recent sectors without pipeline products we added our own TESScut photometry (Brasseur 2019).

A box of the catalogued depth and duration was slid along the data. A repeat needs SNR ≥8, a depth of 0.5–2 times the catalogued value, and at least 80% coverage. Periods were then tested against every predicted epoch, and a period is rejected if one falls on flat data. The red-noise factor is measured on a baseline of at least twice the duration, so long eclipses on giants are not misjudged. For eccentric binaries, primary and secondary eclipses were assigned jointly, and every predicted eclipse in the data had to be seen.

Final periods come from trapezoid fits to each eclipse and a linear ephemeris. Three blind analyses with independent code, starting from the catalogue epoch alone, re-derived every period in Table 1. Apart from EPIC 202126877, which shows timing variations, they agree with ours within 2σ. An adversarial review found pipeline problems that we fixed before the final run. As a validation, Sanders 617 in M67 comes out at 62.59466±0.00018 d. Sandquist 2021 give 62.59484±0.00003 d.

Before claiming a period, we checked each star against ExoFOP, VizieR (including VSX), SIMBAD, the Villanova K2 EB catalogue and Gaia DR3 (Gaia 2023,Arenou 2023).

Results

Of the 144 events, 62 show at least one candidate repeat, and 16 have a period that explains two or more. Of these, 14 are in Table 1. EPIC 202137580 is already well catalogued, and the giant EPIC 212694013 keeps over ten aliases (973.79/N d). A fifteenth system, HD 220967, has primary and secondary eclipses of similar depth that no single period fits; a joint ephemeris resolves it. Events with a single repeat were not vetted systematically, so the table is not complete.

TOI-5560.01 (EPIC 211924561). SPOC alerted this candidate from a single transit in Sector 46; after a second in Sector 72, the maximum period was 702.47 d. The K2 C5 transit and a C16 transit fit one linear ephemeris with the TESS pair. The C16 transit is masked in the EVEREST and K2SFF products but clear in K2 PDCSAP and raw SAP data. The cycle counts (7, 17, 22) share no common factor, and flat data exclude every shorter period except half. This gives P=140.49392±0.00014 or 70.24696±0.00007 d, with T_0=BJD_{TDB} 2457179.8979±0.0014. The transit is V-shaped (≈2.3 ppt, 3.0 h) and far shorter than a central transit across this F star (12–15 h), so the orbit is grazing, eccentric, or both. Gaia DR3 epoch radial velocities are not variable (p=0.08, 22 transits). Their scatter

implies M_{2sini} □0.1 M_{sun} for a circular orbit. Both periods predict a transit at BJD_{TDB} 2461394.716±0.004 (2026 December 20, 05:03 UTC). A transit at 2461464.963 (2027 February 28) would confirm 70.25 d. The 100-d placeholder period in the Villanova catalogue and a 31.27-d SPOC TCE (Valizadegan 2025) are both excluded by the four transits.

EPIC 211503363. This single-transit planet candidate of Kovacs 2020 transits again in Sectors 45 and 72 (3.4–3.8 ppt, 23 h). This leaves P=122.9052±0.0006 or 61.4526±0.0003 d; the detrended QLP flux removes both TESS events, but QLP SAP flux and our own photometry show them. Its host is evolved: Gaia DR3 gives R_□=3.6 R_{sun} and logg=3.46. The companion is therefore ≈2.1–2.4 R_J. That is larger than a planet or brown dwarf on a non-irradiated orbit, and matches an M dwarf. Kovacs 2020 lists R_p=0.260 in a column labelled Jupiter radii, but √(0.004019)×4.098 R_{sun}=0.260 R_{sun} (2.5 R_J), so the value is in solar radii. His BLS period of 73.91 d is a formal value close to the data span, not one of our aliases.

Eclipsing binaries. HD 220967 (G=9.4, F7/8 V) shows two eclipse types, 37% and 30% deep, in K2 C12 and TESS. The secondary falls at phase 0.867, so ecosω≈0.59 and e≥0.58. It lasts 1.9 times as long as the primary, which suggests e≈0.66. It is a bright, long-period benchmark for spectroscopy. For EPIC 211892898, a red giant, the eclipse ephemeris matches the Gaia DR3 SB1 orbit (72.53±0.03 d). The eclipses fall at the predicted primary conjunctions within 1.8σ. Libralato 2016 had noted its eclipses. For EPIC 211489484 our ephemeris excludes the 106.94-d period of a SPOC TCE (Valizadegan 2025).

EPIC 202126877 is a known 21.78-d EB (IJspeert 2024), but the K2 data change the picture. The K2 eclipse lies on cycle -124 of the TESS-only ephemeris, 2 h early, consistent with the eclipse-timing scatter of 10–25 min seen in TESS. It is only 6.2% deep in both EVEREST and K2SFF, and K2 shows no secondary (<0.1%). In our own TESS photometry with a fixed aperture, the primary is 20% deep in 2021, 26% in 2023 and 32% in 2025. Over the same time the secondary grows from 4% to 11%. Dilution cannot change the secondary-to-primary ratio (0.21 to 0.43), and the K2 aperture is uncrowded (CROWDSAP 0.999). A precessing orbit, perhaps driven by a third body, is a plausible cause.

For the C0 binaries, our 34.7456-d period for EPIC 202072917 agrees with the QLP period of IJspeert 2024 and a SPOC TCE (Valizadegan 2025). Our 22.9041-d period for EPIC 202071902 agrees with Guo 2025 and is 0.045 d (3σ) from the SB2 period of Kovalev 2024. Their VSX periods (3.842 and 8.697 d) come from a machine-learning classification (Armstrong 2016). Those periods would put several eclipses in every TESS sector, which is not seen.

Table 1. Periods from K2 revisits and TESS.

EPIC	TIC	Events^e	Period (d)	Prior period
211924561	238606535	C5, C16; S46, S72	140.49392(14) or 70.246961(70)	SPOC TOI: ≤
211503363	20444116	C5; S45, S72	122.90522(64) or 61.45261(32)	none (single;
211411112	437034781	C5, C16; S46	62.59466(18)	62.59484(3) d
246194159	301328232	C12; S70, S92	73.2451778(74)	none
211489484	6951741	C5, C18; S7, S46	82.258438(52)	TCE 106.94 d
211892898	184895645	C5, C16, C18	72.5089(14)	Gaia SB1 72.
248811085	349987241	C14; S46, S72	49.250131(13)	none
246089124	49603732	C12; S70, S92	73.962413(20)	none
246696483	436612436	C13; S43, S70	182.24526(18) or 91.122630(90)	none
211390677	366568535	C5, C18; S45	184.3109(11) or 92.15543(55)	none
247967714	268396693	C13; S44, S71	185.626676(21)/n, n=1–3	none
212012030	14394716	C5, C18; S71	220.321779(57)/n, n=1–5	none
202126877	86491361	C0; S43–45, S71–72	21.775	21.78 d(b)
202072917	308255000	C0; S6, S43, S45	34.7455865(70)	34.744 d(b), 3
202071902	57051340	C0; S44, S71	22.9041282(30)	22.859(15) d^

Uncertainties in the last digits in parentheses. Except for the primary–secondary systems, a period twice as long with identical eclipses cannot be excluded photometrically. Prior periods: (a) Sandquist 2021; (b) IJspeert 2024; ^cSPOC TCE (Valizadegan 2025); ^dKovalev2024, see also Guo 2025. ^eK2 campaigns (C) and TESS sectors (S) with events used in the fit. ^fOur own TESScut photometry, fixed aperture, 2021–2025.

Conclusion

K2 revisits and TESS turn many K2 single events into periods. Most are eclipsing binaries, but the search also narrows a TESS planet candidate to two periods and identifies a K2 planet candidate as a probable M dwarf. Code, fits and all three blind analyses are at https://github.com/coracle-us/duo-transits.

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Facilities: Kepler (K2), TESS, Gaia

Software: lightkurve (Lightkurve 2018), astroquery (Ginsburg 2019), astrocut (Brasseur 2019)

AI disclosure statement

The author used an AI assistant, Anthropic's Claude (a large language model), under his supervision. It wrote and ran the analysis code, including three blind re-analyses and three adversarial reviews performed by separate AI agents, and drafted and edited the text of this note. The author reviewed and edited the text, checked the numbers against the saved analysis outputs, and takes full responsibility for the content. Every reference was checked against ADS, arXiv, Crossref or VizieR records. The note contains no AI-generated figures.

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